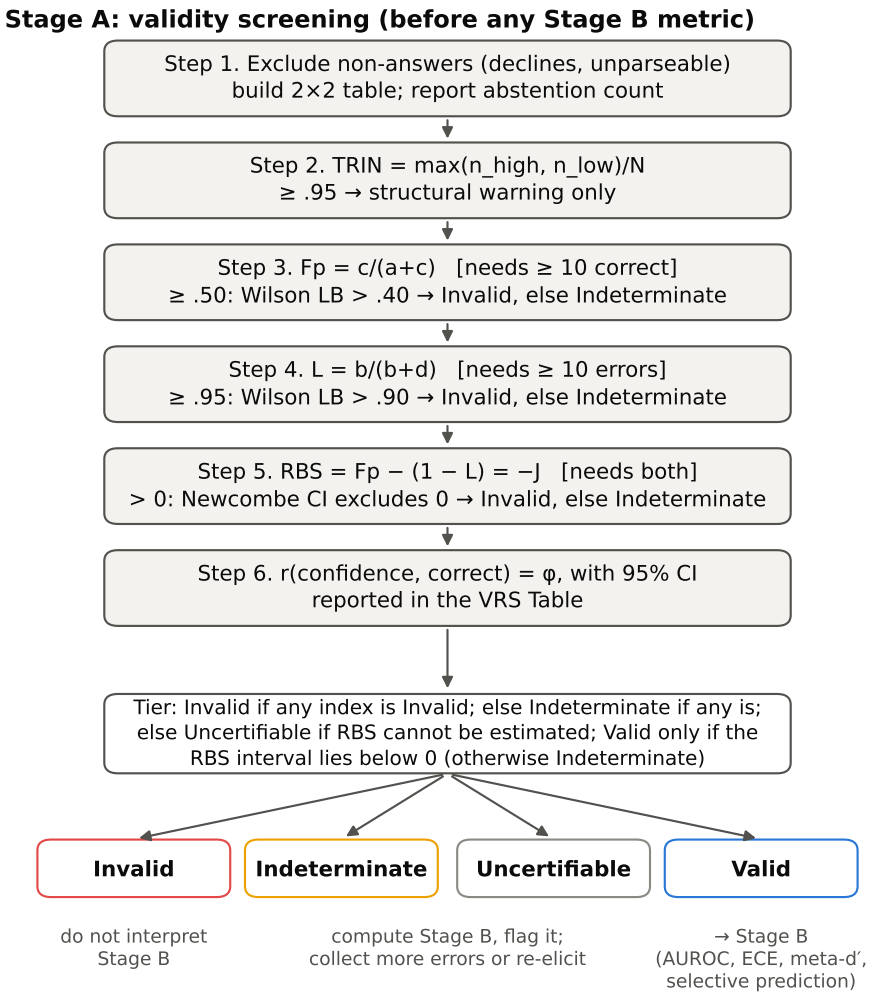
Stage A: validity screening (before any Stage B metric)
Step 1. Exclude non-answers (declines, unparseable)
build 2×2 table; report abstention count
Step 2. TRIN = max(n_high, n_low)/N
≥ .95 → structural warning only
Step 3. Fp = c/(a+c) [needs ≥ 10 correct]
≥ .50: Wilson LB > .40 → Invalid, else Indeterminate
Step 4. L = b/(b+d) [needs ≥ 10 errors]
≥ .95: Wilson LB > .90 → Invalid, else Indeterminate
Step 5. RBS = Fp − (1 − L) = −J [needs both]
> 0: Newcombe CI excludes 0 → Invalid, else Indeterminate
Step 6. r(confidence, correct) = φ, with 95% CI
reported in the VRS Table
Tier: Invalid if any index is Invalid; else Indeterminate if any is;
else Uncertifiable if RBS cannot be estimated; Valid only if the
RBS interval lies below 0 (otherwise Indeterminate)
Invalid
Indeterminate
Uncertifiable
Valid
do not interpret
Stage B
compute Stage B, flag it;
collect more errors or re-elicit
→ Stage B
(AUROC, ECE, meta-d′,
selective prediction)